# Decision Making is Hard

CSU, Chico Math 314

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# outline

Recap

Decision Making is hard

Multiple Comparisons – dangerous implications
- idea
- problem
- example
- take away

# outline

# recap, ANOVA

ANOVA compares the means of $k$ groups in the hypothesis

$$\begin{aligned} H_0 &: \quad \mu_1 = \mu_2 = \ldots = \mu_k \\ H_A &: \quad \text{at least one mean is different} \end{aligned}$$

The test brought us a new statistic with a new standard error.

# outline

# Hypothesis Testing, the decision

We conclude a formal hypothesis test by making a decision between $H_0$ and $H_1$. But did we decide correctly?

# Hypothesis Testing, decision errors

We essentially made a choice, but our choice could be correct or incorrect.

| | fail to reject $H_0$ | reject $H_0$ |
|---|---|---|
| $H_0$ true | Correct | Type I Error |
| $H_A$ true | Type II Error | Correct |

# Hypothesis Testing, errors

### Type I Error

A Type I (1) Error is rejecting the null hypothesis when $H_0$ is actually true.

### Type II Error

A Type II (2) Error is failing to reject the null hypothesis when the alternative is actually true.

# Hypothesis Testing, error trade offs

Consider the following example: how could we reduce the Type II Error rate in US courts? What influence would this have on the Type I Error rate?

- Lower Type II Error by convicting more (guilty) people.
- What then of wrongful convictions, i.e. Type I Error rates?

A trade off exists when considering both types of error.

# outline

# Multiple Comparisons

Running multiple tests on pairs of variables is called **multiple comparisons** and increases the Type I Error rate. For instance

- Suppose you have data across a categorical variable with $k (= 8)$ levels, and you want to compare the means across groups. What do you do?
- DON'T do this ... immediately compare all pairwise combinations of the $k$ groups. That would result in 28 tests/confidence intervals.

# Multiple Comparisons, the problem

If 28 tests at $\alpha = 0.05$ took place, then instead of the planned 5% error, there is a

```
> 1-(.95^28)
## [1] 76.2173
```

percent chance you made an incorrect decision.

# Multiple Comparisons, solution

The **Bonferroni correction** uses a smaller significance, calculated from the significance level you originally chose.

## Bonferroni correction

Given a pre-chosen significance level $\alpha$, the Bonferroni correction specifies a new significance level to use

$$\alpha^* = \alpha / K,$$

where $K = \binom{k}{2}$ and $k$ is the number of groups.

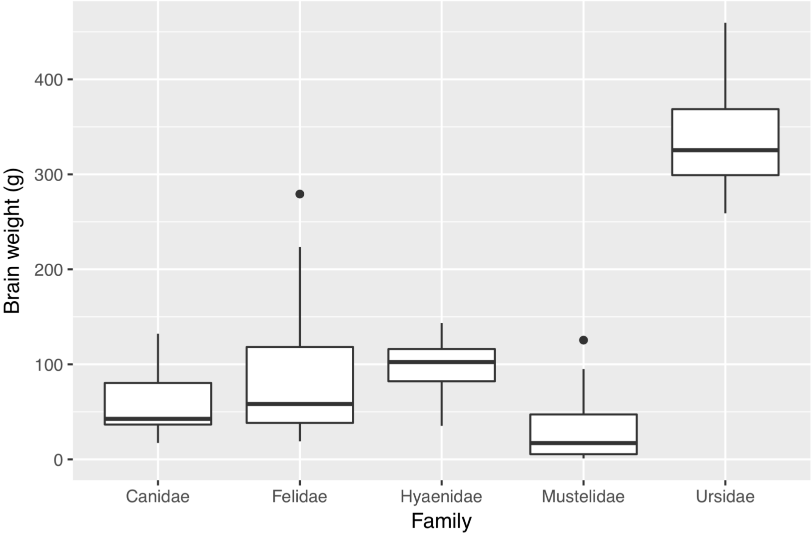
# ape::carnivora, average brain weight

Suppose we want to compare morphological characteristics across some carnivorous families.

```
suppressMessages({library(dplyr)
    library(ape)
    data(carnivora)})
carnivs <- filter(carnivora,
                  !(Family %in% c("Ailuridae",
                                  "Procyonidae",
                                  "Viverridae")))
p <- ggplot(data=carnivs, aes(Family, SB)) +
    geom_boxplot() +
    labs(y="Brain weight (g)")
```

# `ape::carnivora`, average brain weight

First things first . . . plot the data!



# `ape::carnivora`, average brain weight

Set up hypothesis test.

$$\begin{aligned} H_0 &: \quad \mu_c = \mu_f = \mu_h = \mu_m = \mu_u \\ H_A &: \quad \text{at least one mean is different.} \end{aligned}$$

with $\alpha = 0.01$.

# ape::carnivora, average brain weight

Fit the model.

```
fit <- lm(SB~Family, data=carnivs)
anova(fit)
```

```
## Analysis of Variance Table
##
## Response: SB
##           Df Sum Sq Mean Sq F value
## Family     4 354999   88750  37.333
## Residuals 70 166406    2377
##              Pr(>F)
## Family    < 2.2e-16 ***
## Residuals
## ---
## Signif. codes:
##   0 '***' 0.001 '**' 0.01 '*' 0.05
##   '.' 0.1 ' ' 1
```

# `ape::carnivora`, average brain weight

Because the p-value is tiny, we reject $H_0$ in favor of the alternative. We can gather from our initial plot which mean is different from the rest, but we still need to use the Bonferroni correction if we want to investigate all pairwise comparisons.

# ape::carnivora, average brain weight

Bonferroni correction implemented across $k$ groups.

```
# Bonferroni correction
(k <- length(unique(carnivs[,"Family"])))

## [1] 5

K <- choose(k, 2)
(astar <- 0.01/K) # new significance level

## [1] 0.001
```

# ape::carnivora, average brain weight

Now we can make pairwise comparisons. We have to subset the data, again, down to two families of immediate interest.

```
carns1 <- filter(carnivs,
                  Family %in% c("Hyaenidae", "Ursidae"))
t1 <- t.test(SB~Family, data=carns1,
             conf.level=1-astar)
```

# ape::carnivora, average brain weight

Here, we have just Hyaenidae and Ursidae.

```
t1

##  Welch Two Sample t-test
##
## data:  SB by Family
## t = -5.1343, df = 4.5629, p-value =
## 0.004747
## alternative hypothesis: true difference in means
## is not equal to 0
## 99.9 percent confidence interval:
##  -605.2959  112.4459
## sample estimates:
## mean in group Hyaenidae
##                  95.900
##   mean in group Ursidae
##                 342.325
```

# ape::carnivora, average brain weight

Or we can subset original data again down to two different families.

```
carns2 <- filter(carnivs,
                 Family %in% c("Canidae", "Felidae"))
t2 <- t.test(SB~Family, data=carns2,
               conf.leve=1-astar)
```

# ape::carnivora, average brain weight

Just Canidae and Felidae.

```
t2

##
##  Welch Two Sample t-test
##
## data:  SB by Family
## t = -1.5525, df = 27.639, p-value =
## 0.1319
## alternative hypothesis: true difference in means is not
## 99.9 percent confidence interval:
##  -94.84829  38.55940
## sample estimates:
## mean in group Canidae
##              59.45556
## mean in group Felidae
##              87.60000
```

# Quick Note on ANOVA

It is possible to reject the null hypothesis using ANOVA and then to not subsequently identify differences in the pairwise comparisons. This does not invalidate the ANOVA conclusion. It only means we have not been able to successfully identify which groups differ in their means.

This can happen if you have many pairwise comparisons to make, or if the groups are on the smaller side (for instance, Ursidae has only 5 individuals).

# Multiple Comparisons, take away

Upfront multiple comparisons inadvertently increases Type I Error rates. In the real world, you'll likely be in one of two possible positions:

- Formal modelling
  - Need to make multiple comparisons with one test, then use ANOVA
  - if you reject $H_0$: Bonferroni correction[1] on $\alpha$ and make pairwise comparisons
- Prediction
  - use Bonferroni correction up front

[1] Bonferroni is an easy correction to use, but is just one of many, e.g. `?TukeyHSD`.